ANSWER KEY — Integration Lesson 27 Test: FTC Part 2

For the grader · full working shown · give credit for correct method with small arithmetic slips · justification questions need the SIGN reasoning stated, not just the answer

1. Find $\frac{d}{dx} \int_3^x t^4 \, dt$.

> **Work:** FTC Part 2: the derivative of an accumulation function is the inside function with t replaced by x. The lower limit 3 only shifts g by a constant, so it plays no role. **Answer: x^4.**

2. Find $\frac{d}{dx} \int_0^x \sqrt{1 + t^2} \, dt$.

> **Work:** straight FTC 2 — no antiderivative needed (and none exists in simple form). Replace t with x. **Answer: $\sqrt{1 + x^2}$.** (Deduct if the student tried to "integrate first" and got tangled — the whole point is that you don't.)

3. Let $g(x) = \int_5^x \cos t \, dt$. Find $g(5)$ and $g'(x)$.

> **Work:** $g(5) = \int_5^5 \cos t \, dt = 0$, because an integral from a number to itself has zero width — nothing has accumulated at the starting line. $g'(x) = \cos x$ by FTC 2. **Answer: $g(5) = 0$ and $g'(x) = \cos x$.**

4. What is wrong with $g(x) = \int_0^x x^3 \, dx$? Write it correctly.

> **Work:** the letter x is doing two jobs at once: it is the fixed upper limit (where to stop) AND the sweeping integration variable (which must run from 0 up to the stopping point). One letter cannot both stand still and move, so the expression is ambiguous. Correct: **$g(x) = \int_0^x t^3 \, dt$** — t is a dummy variable that sweeps and vanishes (like the index k in sigma notation); x is the real input. **Grader note:** full credit requires BOTH the "two jobs / dummy variable" explanation and the corrected notation.

5. Find $\frac{d}{dx} \int_0^{x^2} \cos t \, dt$.

> **Work:** the upper limit is x^2, a function of x → chain rule. Plug the limit into f: $\cos(x^2)$; multiply by its derivative: $2x$. **Answer: $2x \cdot \cos(x^2)$.** (Common error: $\cos(x^2)$ alone — that misses the chain-rule factor and earns half credit at most.)

6. Find $\frac{d}{dx} \int_2^{x^3} \sqrt{1 + t} \, dt$.

> **Work:** plug the upper limit x^3 into f: $\sqrt{1 + x^3}$; multiply by $\frac{d}{dx}(x^3) = 3x^2$. The lower limit 2 is irrelevant. **Answer: $3x^2 \cdot \sqrt{1 + x^3}$.**

7. Find d/dx $\int_x^4 (t^3 + 1)\,dt$.

Work: the variable is in the LOWER limit. Flip the limits and the sign: $\int_x^4 = -\int_4^x$. Then FTC 2: derivative = $-(x^3 + 1)$. Sense check: as x grows, the interval [x, 4] shrinks, so the total is shrinking (for positive f) — negative derivative is right. **Answer: $-(x^3 + 1) = -x^3 - 1$.** (Common error: forgetting the minus sign.)

8. $g(x) = \int_0^x (t^2 - 9)\,dt$ for $x \geq 0$: find g(0), g′(x), g′(4), and where g is decreasing (justify).

Work: $g(0) = \int_0^0 = 0$ (zero-width integral). $g'(x) = x^2 - 9$ by FTC 2. $g'(4) = 16 - 9 = 7$. Decreasing: $g'(x) = x^2 - 9 = (x - 3)(x + 3) < 0$ for $0 \leq x < 3$. **Answer: $g(0) = 0$; $g'(x) = x^2 - 9$; $g'(4) = 7$; g is decreasing on (0, 3) because $g'(x) = x^2 - 9$ is negative there. Grader note:** the justification must mention the SIGN of g′ (or of f) — "because the graph goes down" earns nothing on the AP.

9. $g(x) = \int_1^x 3t^2\,dt$: tangent line to g at x = 2.

Work: Point: by FTC 1 with antiderivative t^3, $g(2) = 2^3 - 1^3 = 8 - 1 = 7 \rightarrow$ point (2, 7). Slope: $g'(x) = 3x^2$ by FTC 2, so $g'(2) = 12$. Line: $y = 7 + 12(x - 2)$. **Answer: $y = 7 + 12(x - 2)$, i.e. $y = 12x - 17$.** (Either form is full credit; point-slope form is fine.)

10. A student wrote d/dx $\int_0^{3x} \sin t\,dt = \sin(3x)$. Find and fix the error.

Work: the upper limit is 3x, not plain x, so the chain rule requires multiplying by d/dx(3x) = 3. The student forgot the chain-rule factor. Correct: $\sin(3x) \cdot 3$. Check the long way: $\int_0^{3x} \sin t\,dt = 1 - \cos(3x)$, whose derivative is $3\sin(3x)$. ✓ **Answer: missing chain-rule factor; correct derivative is $3\sin(3x)$. Grader note:** require the one-sentence explanation naming the chain rule (or "derivative of the upper limit"), not just the fixed answer.